Lecture 1

Crisis in Mathematics

The subject of (real) analysis was born out of an unresovable mathematical crisis in the 19th century. It fundamentally came down to the lackadaisical use of infinity and infinitesmals found within calculus. We will explore a few of these problems here.

Problem 1: Infinite Summations / Series

We are familiar with (finite) summations such as $1+2$, $2+3$ etc.

An infinite series can be understood as a sum with no end. For example,

$$1+2+3+4+\cdots.$$

The '...' is used to mean 'and so on'.

It turns out some summations get closer and closer to single values, or even functions, the more terms are added. It is then tempting, and it turns out useful,

to treat these infinite series as infinite summations. That is, an infinite process of usual addition that reaches some value / function.

For example, the sum of functions,

$$1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \dots$$

This gets closer and closer to the function e^{-x^2}.

It seems reasonable to say that

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \dots$$

Why is this useful? In this case, integrating e^{-x^2} is very difficult but realising it as a power series, and assuming integration is linear over this series, allows us to integrate.

(Linear $\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$)

That is, we can say

$$\int e^{-x^2}\,dx = x - \frac{x^3}{3} + \frac{x^5}{5(2!)} - \frac{x^7}{7(3!)} + \cdots$$

Are all <u>convergent</u> infinite series, really just infinite summations?

Consider the infinite series:

$$1 - 1/2 + 1/3 - 1/4 + 1/5 - 1/6 + \cdots = \ln(2).$$

If this really is a summation, then surely

$$1 + 1/3 - 1/2 + 1/5 + 1/7 - 1/4 + \cdots \approx 1.038 \neq \ln(2)$$

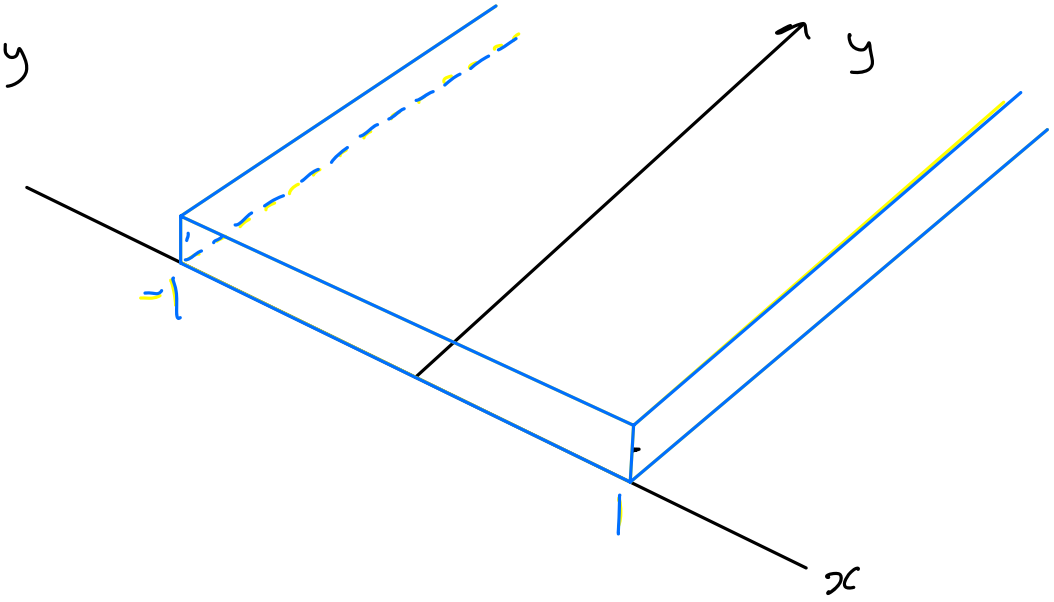
should come to the same value.

What is going on? Why is it that some convergent infinite series can be treated as infinite summations but others can't?

Problem 2 - Fourier Series

Fourier series arose as a result of the study of propagation of heat in solid bodies. That is, how does heat flow through solid bodies. Consider the

simple body



No heat is lost to surroundings and the long sides of the sheet have zero temperature. So,

$$T(-1, y) = T(1, y) = 0 \quad \text{for } y > 0.$$

So if there is an initial temperature distribution along $y=0$, what is the temperature throughout the body?

Assuming the initial temperature distribution is

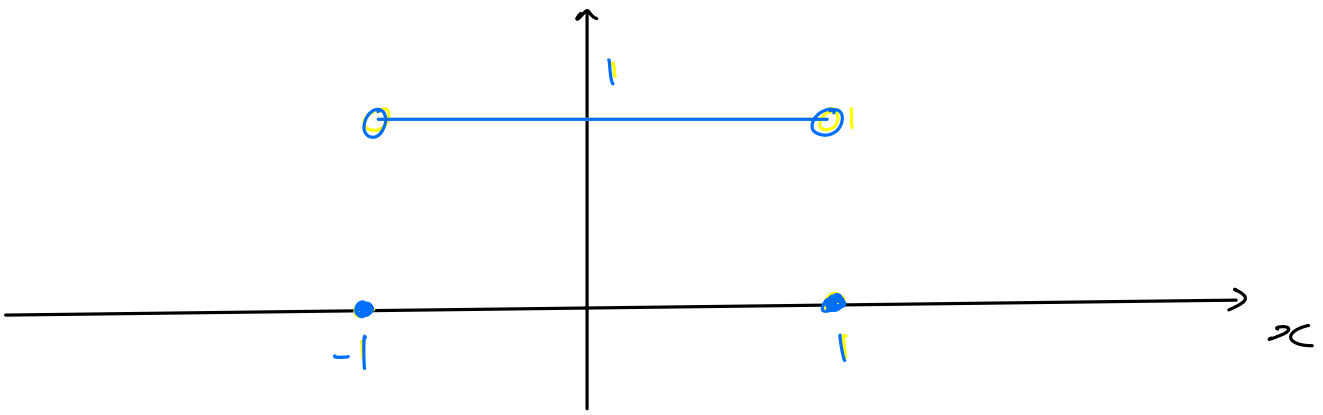
$$f(x) = a_1 \cos\left(\frac{\pi x}{2}\right) + a_2 \cos\left(\frac{3\pi x}{2}\right) + \cdots + a_n \cos\left(\frac{(2n-1)\pi x}{2}\right)$$

where $a_i \in \mathbb{R}$ then

$$T(x,y) = a_1 e^{-\pi y/2} \cos\left(\frac{\pi x}{2}\right) + a_2 e^{-\frac{3\pi y}{2}} \cos\left(\frac{3\pi x}{2}\right) + \cdots$$

$$+ a_n e^{-(2n-1)\pi y/2} \cos\left(\frac{(2n-1)\pi x}{2}\right).$$

What if the initial temperature is constant? That is, what if $f(x) = 1$, for $-1 < x < 1$.



Fourier claimed that

$$\frac{4}{\pi}\left(\cos\left(\frac{\pi x}{2}\right) - \frac{1}{3}\cos\left(\frac{3\pi x}{2}\right) + \frac{1}{5}\cos\left(\frac{5\pi x}{2}\right) + \cdots\right) = 1,$$

for $-1 < x < 1$. This turned out to be correct, and could be physically measured.

If this claim was correct, it would shock the very understanding of infinite series, differentiation, continuity, and even the meaning of function itself. Further, it started mathematicians to question the very number systems they were working with and this led

to a redevelopment of how mathematics was done.

What is a number?

As a representation of physical measurement.

The simplest form of measurement is counting discrete objects.

Defn :: The set of <u>natural numbers</u>, denoted by $\mathbb{N}$, are given by

$$\mathbb{N} := \{1, 2, 3, 4, \dots\}.$$

Continuing with discrete measures, we ask what are the numbers that measure the absence of discrete objects.

Namely, debt.

Defn : The set of <u>integers</u>, denoted by $\mathbb{Z}$, are given by

$$\mathbb{Z} := \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

Moving away from discrete measures, to cts measures, like lengths, we need fractional quantities.

Defn: The set of <u>rational numbers</u>, denoted by $\mathbb{Q}$, is given by

$$\mathbb{Q} := \{ p/q : p \in \mathbb{Z}, q \in \mathbb{N} \}.$$

In addition to these numbers representing measurements, the measurements also have relations between one another. For example, 1 apple plus another equals two.

That is, $\mathbb{Q}$ has on it two binary operations, namely $+$ (addition), $(\cdot)$ multiplication alongside a relation $<$ (ordering).

From now on, we view $\mathbb{Q}$ as a single algebraic object with all these operations.

$$\mathbb{Q} := (\mathbb{Q}, +, \cdot, <).$$